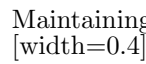


Free Energy and Chemical Thermodynamics If you conduct any chemical experiment; for example, drop an aluminum

Maintaining the constant volume condition is impossible; because our substances are constantly changing in chemical reactions.  How much energy required to create steam from out of nothing? The answer is quite

Like entropy, we actually do not care much about the enthalpy  $H$  of a stable system; it is not so meaningful when I say "A

I have always thought of enthalpy as a representation of the state of all substances, in terms of energy. Look at the left side

Conversely, the surrounding environment needs to **provide** energy to decompose  $H_2O$  into  $H_2$  and  $O_2$ ; this is called an **end**

The enthalpy of vaporization of water at  $100^\circ\text{C}$  is  $40.63 \text{ (kJ/mol)}$ . How much energy is needed to create space for steam? From Definition , we see that:

The amount of work  $PV$  can be calculated by using ideal gas law: